

This question paper contains 4 printed pages.

Your Roll No.

No. of Paper : 91
Unique Paper Code : 32351302
Name of the Paper : Group Theory - I
Name of the Course : B.Sc. (Hons.) Mathematics
Semester : III
Duration : 3 hours
Maximum Marks : 75

*(Write your Roll No. on the top immediately
on receipt of this question paper.)*

*Attempt any two parts from each question.
All questions are compulsory.*

- (a) Define a group. Give an example of:
- (i) an abelian group consisting of eight elements,
 - (ii) a non-abelian group consisting of six elements,
 - (iii) an infinite abelian group, and
 - (iv) an infinite non-abelian group.
- (b) Show that the set $\{5, 15, 25, 35\}$ is a group under multiplication modulo 40. What is the identity element of this group? Find the inverse of each element.
- (c) Prove that the intersection of an arbitrary family of subgroups of a group G is again a subgroup of G . What can you say about the union of two subgroups? Justify your answer.

2×6=12

P. T. O.

2. (a) (i) Prove that in $(\mathbb{Z}, +)$, the group of integers under addition, every non-zero element is of infinite order.
- (ii) Let G be a group and $a \in G$. If $|a| = n$ and k is positive divisor of n , then prove that $|a^{n/k}| = k$.
- (b) Prove that the order of a cyclic group is equal to the order of its generator.
- (c) Define a cyclic group. If $G = \langle a \rangle$ is a finite cyclic group of order n , then prove that the order of any subgroup of G is a divisor of n , and for each positive divisor k of n , G has exactly one subgroup of order k namely, $\langle a^{n/k} \rangle$.

$$2 \times 6.5 = 13.5$$

3. (a) Prove that if the identity permutation $\varepsilon = \beta_1 \cdots \beta_r$ where the β 's are 2-cycles then r is even.

(b) Show that for $n \geq 3$, $Z(S_n) = \{I\}$.

(c) Prove that:

- (i) a group of prime order has no proper, non-trivial subgroup. State its converse. Is it true?
- (ii) a group of prime order is cyclic and any non-identity element can be taken as its generator.

$$2 \times 6 = 12$$

4. (a) Let G be a finite group of permutations of a set S . Then prove that for any i from S :

$$|G| = |\text{orb}_G(i)| |\text{stab}_G(i)|.$$

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(b) (i) Prove that the center $Z(G)$ of a group G is a subgroup of G and is normal in G . ✓

(ii) If H is a subgroup of G such that H is contained in the center $Z(G)$, then prove that H is a normal subgroup of G . Is the converse true? Justify your answer.

(c) Let N be a normal subgroup of a group G and let H be a subgroup of G . If N is a subgroup of H , prove that H/N is a normal subgroup of G/N if and only if H is a normal subgroup of G . $2 \times 6.5 = 13$

3. (a) Let $\mathbf{C}$ be the complex numbers and:

$$\mathbf{M} = \left\{ \begin{bmatrix} a & -b \\ b & a \end{bmatrix} : a, b \in \mathbb{R} \right\}.$$

Prove that $\mathbf{C}$ and $\mathbf{M}$ are isomorphic under addition and $\mathbf{C}^* = \mathbf{C} \setminus \{0\}$ and $\mathbf{M}^* = \mathbf{M} \setminus \{0\}$ are isomorphic under multiplication.

(b) Prove that an infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$. Hence show that every subgroup of an infinite cyclic group is isomorphic to the group itself.

(c) Let G be a group of permutations. For each σ in G , define

$$\text{sgn}(\sigma) = \begin{cases} 1, & \text{if } \sigma \text{ is an even permutation,} \\ -1, & \text{if } \sigma \text{ is an odd permutation.} \end{cases}$$

Prove that sgn is a homomorphism from G to $\{1, -1\}$.
What is the kernel? $2 \times 6 = 12$

P. T. O.

6. (a) Let ϕ be a homomorphism from a group G to a group $\tilde{G}$. Let g be an element of G . Then:

(i) $\phi(g^n) = \phi(g)^n$ for all $n \in \mathbb{Z}$.

(ii) ϕ is one-one if and only if $\ker(\phi) = \{e\}$, where e is the identity of G .

(b) State and prove the First Isomorphism Theorem.

(c) (i) Suppose ϕ is a homomorphism from $U(30)$ to $U(30)$ and $\text{Ker}(\phi) = \{1, 11\}$.

If $\phi(7) = 7$, find all elements of $U(30)$ that map to 7.

(ii) Let G be a group. Prove that the mapping $\phi(g) = g^{-1}$, for all $g \in G$, is an isomorphism from G onto G if and only if G is Abelian.

$$2 \times 6.5 = 13$$